

CMB Compatibility and the Dissolution of the Mass Hierarchy in the Elastic Vacuum

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What If the Vacuum Gravitates Locally?

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Abstract

The preceding paper in this programme (Paper 6, Kriger 2026a) derived the Starobinsky potential $V(\psi)$ from the elastic properties of spacetime, proved the flatness feedback parameter $\beta > 0$ in the quasi-static limit, and identified a 10^{55} mass hierarchy between the inflationary scalaron ($m \sim 10^{13}$ GeV) and the late-time screening field ($m_\psi \sim 10^{-33}$ eV) as the programme’s deepest unsolved problem. This paper delivers two results. First, we show that the mass hierarchy dissolves within standard effective field theory: the full gravitational effective action $S_{\text{eff}} = \int d^4x \sqrt{-g} [c_0 \Lambda + c_1 R + c_2 R^2 + c_3 R^3 + \dots]$ supports different effective degrees of freedom at different background curvatures. At inflationary curvature $R \sim M^2$, the R^2 correction dominates and yields a heavy scalaron, which is *derived* from the action via conformal transformation. At late-time curvature $R \sim H_0^2$, the screening field ψ with $m_\psi \sim H_0$ is *consistent with* the infrared sector of the effective action, though not yet derived from it. These are distinct excitation branches of the same elastic medium—analogueous to optical phonons and seismic waves in a solid—not a single field required to span 55 orders of magnitude. Second, we establish compatibility with the cosmic microwave background (CMB). At recombination ($z \approx 1089$), vacuum energy screening is complete, the screening field ψ is frozen in its quasi-static configuration, and the modified Friedmann equations reduce to standard Λ CDM. We show that ψ -perturbations are suppressed at recombination by a factor $(H_0/H_{\text{rec}})^2 \sim 10^{-10}$, leaving the acoustic peak structure unaffected at leading order. A full dynamical solution of the coupled (ψ, a, K) system confirms $\beta > 0$ beyond the quasi-static approximation, with the quantitative flatness evolution consistent with the Planck bound $|\Omega_K| < 0.001$. Heuristic scaling estimates suggest an enhanced integrated Sachs–Wolfe (ISW) effect from void evolution as a qualitative indication of where the framework first departs from Λ CDM; confirmation requires implementation in a Boltzmann code.

Keywords: elastic vacuum, CMB compatibility, mass hierarchy, effective field theory, vacuum energy screening, flatness problem, cosmological perturbations, ISW effect, Starobinsky inflation

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1 Introduction

The elastic vacuum programme [1] develops the hypothesis that quantum vacuum energy gravitates locally—within bound structures—while being screened from cosmological expansion. Paper 5 [1] established the dynamical framework: a screening scalar field ψ coupled to the matter trace $T^\mu{}_\mu$ via a parameter ξ , producing scale-dependent gravitational screening and a flatness feedback mechanism conditional on a positive feedback parameter $\beta > 0$. Paper 6 [2] derived the scalar field potential $V(\psi)$ from the elastic properties of spacetime—the Starobinsky potential [3], arising from the Einstein–Hilbert action supplemented by its leading quantum correction (R^2)—and proved $\beta > 0$ in the quasi-static limit from the convexity of $V(\psi)$.

Paper 6 left three open questions:

- (i) The **mass hierarchy**: the inflationary scalaron has mass $m \sim M \sim 1.3 \times 10^{13}$ GeV, while the late-time screening field requires $m_\psi \sim H_0 \sim 10^{-33}$ eV—a ratio of $\sim 10^{55}$, identified in Paper 6, §6.5, as the programme’s “deepest unsolved problem.”
- (ii) **CMB compatibility**: the analytical argument that the elastic vacuum reproduces the observed CMB acoustic peak structure, promised in Papers 5 and 6 but not yet delivered.
- (iii) **Full dynamical β** : numerical confirmation of $\beta > 0$ beyond the quasi-static approximation, including the coupled evolution of (ψ, a, K) , promised in Paper 6, §8.

This paper addresses all three. We show that the mass hierarchy is not a problem to be solved but a misframing to be dissolved: the effective field theory (EFT) of gravity naturally supports different excitation modes at different curvature scales, and no single field is required to interpolate between them. We establish CMB compatibility by demonstrating that at recombination, the screening field is frozen and its perturbations are negligible compared to Planck error bars. And we present a full numerical solution confirming $\beta > 0$ dynamically.

We are explicit about an asymmetry in the current state of the programme that the dissolution argument clarifies but does not eliminate: the inflationary scalaron is *derived* from the gravitational effective action (Paper 6, §5), while the late-time screening field is *postulated* on phenomenological grounds (Paper 5) and shown here to be *consistent with* the EFT framework. A first-principles derivation of the screening mode from the infrared sector of the effective action remains an open problem.

The paper is organized as follows. Section 2 maps the logical structure. Section 3 dissolves the mass hierarchy. Section 4 analyzes the elastic vacuum at recombination. Section 5 develops the perturbation analysis. Section 6 presents the quantitative CMB analysis. Section 7 gives the full dynamical β . Section 8 identifies predictions. Sections 9–11 provide discussion, limitations, and future directions.

2 Logical Structure of the Argument

The reasoning proceeds through six steps, each building on the previous.

Step 1 (§3): *The mass hierarchy dissolves.* The 10^{55} ratio between inflationary and late-time mass scales does not require explanation as a dynamical hierarchy within a single field. The full gravitational effective action supports different effective degrees of freedom at different background curvatures. The inflationary scalaron (derived from R^2) and the late-time screening mode (phenomenologically motivated, consistent with the EFT) are distinct excitation branches of the same elastic medium. The temporal continuity of the screening field—its state between the end of inflation and the onset of late-time dynamics—is addressed explicitly.

Step 2 (§4): *The elastic vacuum at recombination.* At $z \approx 1089$, the matter density vastly exceeds the effective vacuum contribution. The screening field ψ is frozen in its quasi-static configuration, and the modified Friedmann equations reduce to standard Λ CDM at leading order. The CMB acoustic peaks are preserved by construction.

Step 3 (§5): *Perturbation analysis.* Scalar perturbations of ψ in the recombination-era background are analyzed. The ψ -perturbations are subdominant by a factor $(H_0/H_{\text{rec}})^2 \sim 10^{-10}$, ensuring that the matter power spectrum is unaffected at linear order. A heuristic scaling estimate suggests an enhanced integrated Sachs–Wolfe (ISW) effect from late-time void evolution, though this requires numerical confirmation.

Step 4 (§6): *Quantitative CMB analysis.* The contributions of ψ to the angular power spectrum C_ℓ are computed analytically and shown to be suppressed below Planck error bars across the full multipole range.

Step 5 (§7): *The full dynamical β .* A numerical solution of the coupled (ψ, a, K) system, derived from the Friedmann and scalar field equations without pre-assuming the β -dependent form, confirms $\beta > 0$ throughout the matter-dominated and Λ -dominated eras, with the quantitative flatness evolution consistent with $|\Omega_K| < 0.001$.

Step 6 (§8): *Predictions.* The framework makes specific predictions: a qualitative ISW enhancement from voids (requiring Boltzmann code confirmation), a specific $w(a)$ trajectory, scale-dependent deviations in the matter power spectrum at $k \sim H_0$, and null results ($\mu = \Sigma = 1$, $c_T^2 = 1$) at linear order. A lower bound on the coupling ξ from the requirement that the elastic vacuum reproduce the observed dark energy density is identified.

Synthesis. The six steps establish that the elastic vacuum framework is compatible with all current CMB observations, that the mass hierarchy is dissolved within standard EFT (with the noted asymmetry), and that the flatness attractor is confirmed dynamically.

3 The Mass Hierarchy Dissolved

Paper 6, §6.5, identified the 10^{55} ratio between the inflationary scalaron mass ($m \sim M \sim 1.3 \times 10^{13}$ GeV) and the late-time screening field mass ($m_\psi \sim H_0 \sim 10^{-33}$ eV) as the programme’s “deepest unsolved problem” and stated candidly that the elastic vacuum achieves “unification of interpretation but not unification of dynamics.” That assessment was too pessimistic. It rested on the implicit assumption that both epochs must be described by a *single dynamical field* with a single mass parameter. Once this assumption is examined, the “problem” dissolves—though an asymmetry between what is derived and what is assumed remains, and we are explicit about it.

3.1 Why the Problem Was Incorrectly Posed

The question “how can one field have mass 10^{13} GeV at inflation and 10^{-33} eV today?” presupposes that the inflationary scalaron and the late-time screening mode *are* one field. But the elastic vacuum programme never required this. Paper 5 [1] introduced a screening scalar field ψ with horizon-scale mass $m_\psi \sim H_0$ as the mediator of late-time vacuum energy screening. Paper 6 [2] derived the Starobinsky potential from $R + R^2$ gravity via conformal transformation, producing an inflationary scalaron with mass $m \sim M \sim 10^{13}$ GeV. These two derivations operate at different curvature scales, use different sectors of the gravitational effective action, and produce different effective degrees of freedom.

The unification claim of the programme is not that one field spans 55 orders of magnitude. It is that one *principle*—the intrinsic elasticity of spacetime—governs both epochs. This is a weaker but physically sufficient claim, and it is the one actually supported by the formalism.

3.2 One Principle, Many Modes: The EFT Perspective

The full effective action of gravity, obtained by integrating out matter fields on a curved background, has the form [8, 9]:

$$S_{\text{eff}} = \int d^4x \sqrt{-g} \left[c_0 \Lambda + c_1 R + c_2 R^2 + c_3 R^3 + c_4 R_{\mu\nu} R^{\mu\nu} + c_5 \nabla^2 R + \dots \right], \quad (1)$$

where the coefficients c_n are determined by the field content and the renormalization scale. This action is an infinite series in curvature invariants, valid at scales below the Planck mass.

The crucial observation is that at different background curvatures $\bar{R}$, different terms in (1) dominate the corrections to the Einstein–Hilbert term $c_1 R$:

- **At inflationary curvature** ($\bar{R} \sim M^2 \sim 10^{-6} M_{\text{Pl}}^2$): The R^2 correction dominates. Higher-order terms (R^3 , R^4 , $\nabla^2 R$) are suppressed by $\bar{R}/M_{\text{Pl}}^2 \sim 10^{-6}$ per additional power. Linearizing around this background and performing the conformal transformation yields the Starobinsky scalaron with mass $m \sim M$. This mode does its job—driving ~ 60 e-folds of inflation—then oscillates and decays, reheating the universe. After reheating, the scalaron is gone: it has transferred its energy to radiation and matter.
- **At late-time curvature** ($\bar{R} \sim H_0^2 \sim 10^{-122} M_{\text{Pl}}^2$): The R^2 term is negligible ($c_2 R^2 / c_1 R \sim H_0^2 / M^2 \sim 10^{-116}$). The relevant physics comes from a different sector of the effective action—the coupling between the vacuum’s elastic response and matter, mediated by the screening field ψ with $m_\psi \sim H_0$. This mode is *not* the inflationary scalaron in disguise; it is a different excitation of the gravitational effective action, active at a different curvature scale.

We must be precise about the epistemic status of these two statements. The inflationary scalaron is *derived*: the $R + R^2$ action, via conformal transformation, yields the Starobinsky potential and the scalaron mass $m \sim M$ (Paper 6, §5). The late-time screening field is *postulated*: Paper 5 introduces ψ with $m_\psi \sim H_0$ on phenomenological grounds, and the

present argument shows that such a mode is *consistent with* the EFT framework (1)—the effective action generically supports excitations at different scales—but does not derive $m_\psi \sim H_0$ from the action. The dissolution of the hierarchy consists in showing that no single field must interpolate between the two scales; it does not consist in deriving both modes from first principles.

3.3 The Condensed-Matter Parallel

The situation is standard in condensed-matter physics and requires no exotic mechanism. Consider a crystalline solid:

- At high frequencies ($\sim 10^{12}$ – 10^{13} Hz), the dominant excitations are **optical phonons**—quantized vibrations of the lattice with a characteristic energy gap ~ 10 – 100 meV.
- At low frequencies ($\sim 10^{-2}$ – 10^2 Hz), the dominant excitations are **acoustic/seismic waves**—long-wavelength collective modes with effective “masses” (frequency scales) many orders of magnitude lower.

The ratio of optical phonon frequency to seismic wave frequency can exceed 10^{11} . Nobody regards this as a “hierarchy problem.” Both modes arise from the same elastic medium (the crystal lattice), governed by the same elastic moduli. They are different excitation branches, active at different scales, described by different effective theories obtained by coarse-graining the same microscopic physics.

The gravitational analogue is instructive but imperfect, and we note the disanalogy. In the condensed-matter case, both the high-frequency and low-frequency modes arise from the *same microscopic Hamiltonian* with the *same coupling constants*; the scale separation comes from wavelength ratios within a single set of equations. In the gravitational case, the inflationary scalaron is controlled by the mass scale M (the R^2 coefficient), while the late-time screening field is controlled by separate parameters ξ and m_ψ (the matter coupling and screening mass). The two modes arise from *different sectors* of the effective action with *different parameters*. Moreover, the gravitational scale ratio (10^{55}) exceeds the condensed-matter ratio (10^{11}) by a factor of 10^{44} , which may signal a qualitative rather than merely quantitative difference.

The analogy is therefore limited: it demonstrates that *large scale ratios between excitation branches are physically generic*, but it does not establish that the gravitational case is as benign as the condensed-matter case. The argument against a “hierarchy problem” rests on the EFT structure (§3.2), not on the condensed-matter analogy alone.

3.4 Temporal Continuity: The Screening Field Between Epochs

The dissolution argument establishes that the inflationary scalaron and the late-time screening field are different modes. This raises a question of temporal continuity: if ψ is a different excitation from the scalaron, when does it become dynamically populated, and in what state does it exist during the radiation and matter-dominated eras?

The answer follows from the structure of the action (Paper 5, Eq. (32)):

$$S = \int d^4x \sqrt{-g} \left[\frac{R - 2\Lambda_0}{16\pi G} + \mathcal{L}_m + \mathcal{L}_{\text{vac}} - \frac{1}{2}(\nabla\psi)^2 - V(\psi) - \xi\psi T^\mu{}_\mu \right]. \quad (2)$$

The field ψ is present in the action at all times—it is a degree of freedom of the theory, not a transient excitation like the scalaron. During the radiation era ($T^\mu{}_\mu = 0$ for radiation), the source $\xi T^\mu{}_\mu$ vanishes, and ψ sits at the minimum of its potential $V(\psi)$ with negligible amplitude. During matter domination, the source $\xi\rho_m$ activates, and ψ responds quasi-statically: it is displaced from its minimum by $\delta\psi \approx \xi\rho_m/(m_\psi^2 + \xi^2\rho_m)$, producing the screening effect.

The transition is continuous and automatic. In condensed-matter language: both the “optical phonon” (scalaron) and the “acoustic mode” (ψ) are always present in the spectrum; what changes is which mode carries significant amplitude. The scalaron is highly excited during inflation and decays during reheating. The screening field ψ is in its ground state during radiation domination, begins responding quasi-statically during matter domination when its source ($\xi\rho_m$) becomes nonzero, and reaches its saturated configuration at high matter density. At late times ($z \lesssim 2$), as matter dilutes, ψ becomes dynamically active and drives the screening effects of Paper 5.

The screening field is therefore not “created” at some transition epoch. It is always present; its dynamics are simply irrelevant during the radiation era (no source) and frozen during early matter domination (saturated quasi-static response). It becomes dynamically interesting only when ρ_m drops to values comparable to m_ψ^2/ξ^2 , which occurs at late times.

3.5 Formal Statement

Proposition 3.1 (Dissolution of the mass hierarchy). *Let $S_{\text{eff}}[g_{\mu\nu}]$ be the gravitational effective action (1), valid at scales below M_{Pl} . At a background curvature $\bar{R}$, the spectrum of linearized excitations around $\bar{g}_{\mu\nu}$ depends on $\bar{R}$ through the coefficients c_n and the background geometry. In particular:*

- (a) *At $\bar{R} \sim M^2$ (inflationary epoch), the R^2 sector yields—via the standard conformal transformation—a scalar excitation with mass $m \sim M \sim 10^{13}$ GeV. This mode is **derived** from the effective action.*
- (b) *At $\bar{R} \sim H_0^2$ (present epoch), the matter-coupled sector of the full action (2) **accommodates** a scalar excitation with mass $m_\psi \sim H_0 \sim 10^{-33}$ eV. This mode is postulated on phenomenological grounds (Paper 5) and is **consistent with** the EFT framework, but is not yet derived from the effective action.*

These are different modes of the same effective action evaluated at different backgrounds. No dynamical mechanism is required to bridge the two scales, because there is no single degree of freedom that must interpolate between them. The “hierarchy” was an artefact of the incorrect assumption that one field must serve both epochs. A first-principles derivation of m_ψ from the infrared sector of the gravitational effective action remains an open problem.

Proof. The spectrum of linearized perturbations around a background $\bar{g}_{\mu\nu}$ is determined by the second variation $\delta^2 S_{\text{eff}}/\delta g_{\mu\nu}\delta g_{\alpha\beta}|_{\bar{g}}$. This is a differential operator whose eigenvalues (masses of the propagating modes) depend on $\bar{g}_{\mu\nu}$ and hence on $\bar{R}$. Part (a): at $\bar{R} \sim M^2$, the dominant correction to the Einstein–Hilbert term is $c_2 R^2$, and the conformal transformation yields the Starobinsky scalaron with mass $m^2 = M^2/3$ (Paper 6, §5). This is a derivation. Part (b): at $\bar{R} \sim H_0^2$, the R^2 correction is negligible; the relevant excitation is the screening field ψ of the full action (2), whose mass $m_\psi \sim H_0$ is set by the coupling ξ to T^μ_μ and the background matter density. That such a mode is consistent with the EFT framework follows from the genericity of light scalars coupled to matter in the infrared; that it is *required* by the framework has not been demonstrated. Since $\delta^2 S_{\text{eff}}/\delta g^2$ evaluated at two different backgrounds yields two different eigenvalue problems, the modes need not—and generically do not—have related masses. $\square$

Counterarguments and Responses.

Objection: This dissolves the hierarchy by declaring two different fields. Isn’t that just adding a new field and calling it a resolution?

Response: The late-time screening field ψ was already present in Paper 5—it is not introduced here. The mass hierarchy “problem” arose from the attempt, in Paper 6, to interpret both the inflationary scalaron and ψ as aspects of a single field. What this section demonstrates is that such identification was never required by the formalism. The two fields arise from different sectors of the same effective action at different curvature scales, which is standard EFT physics.

Objection: Proposition 3.1 is asymmetric: part (a) is a derivation, part (b) is an assumption dressed as a result. What has actually been proven?

Response: We agree with this characterization and have stated it explicitly in the proposition. What has been proven is the *negative* result: there is no inconsistency in having both a heavy mode at high curvature and a light mode at low curvature. The positive result—deriving $m_\psi \sim H_0$ from the effective action—remains open. The dissolution of the hierarchy does not require deriving both modes; it requires only showing that no single mode must span both scales.

Objection: If the two modes are unrelated, what remains of the “unified elastic framework”?

Response: The unification is at the level of *principle*, not parameter. Both modes arise from the elastic properties of spacetime as encoded in (1). Both are excitations of the same medium. The programme unifies the physical picture—inflation as elastic recoil, late-time acceleration as elastic relaxation—without requiring a single dynamical field to span all scales.

Section Result and Implications. The mass hierarchy between the inflationary scalaron and the late-time screening field is not a dynamical problem requiring a bridging mechanism. It is a consequence of different background curvatures activating different excitation branches of the same elastic medium. The dissolution is *asymmetric*: the high-curvature mode is derived, the low-curvature mode is assumed and shown to be consistent. This asymmetry is a limitation of the current programme, not of the dissolution argument itself. The “problem” identified in Paper 6, §6.5, was based on the incorrect assumption that both epochs must be described by a single field.

4 The Elastic Vacuum at Recombination

We now show that the elastic vacuum framework reduces to standard Λ CDM at recombination ($z \approx 1089$), ensuring compatibility with the observed CMB acoustic peak structure.

4.1 Background Cosmology at Recombination

The modified Friedmann equation from Paper 5, §6, is:

$$H^2 = \frac{8\pi G}{3} (\rho_m + \rho_r + \rho_\psi) + \frac{\Lambda_0}{3} - \frac{K}{a^2}, \quad (3)$$

where $\rho_\psi = \frac{1}{2}\dot{\psi}^2 + V(\psi)$ is the energy density of the screening field and Λ_0 is the geometric cosmological constant (separated from the QFT vacuum energy by the sequestering mechanism of Paper 5).

At recombination ($z_{\text{rec}} \approx 1089$, $a_{\text{rec}} \approx 9.2 \times 10^{-4}$), the energy densities scale as:

$$\rho_{m,\text{rec}} = \rho_{m,0} a_{\text{rec}}^{-3} \approx 1.3 \times 10^9 \rho_{m,0}, \quad (4)$$

$$\rho_{r,\text{rec}} = \rho_{r,0} a_{\text{rec}}^{-4} \approx 1.4 \times 10^{12} \rho_{r,0}, \quad (5)$$

$$\rho_{\Lambda,\text{rec}} = \frac{\Lambda_0}{8\pi G} \approx \rho_{\Lambda,0}, \quad (6)$$

where subscript 0 denotes present-day values. With $\Omega_m = 0.315$, $\Omega_r = 9.1 \times 10^{-5}$, and $\Omega_\Lambda = 0.685$:

$$\frac{\rho_{\Lambda,\text{rec}}}{\rho_{m,\text{rec}}} \sim \frac{\Omega_\Lambda}{\Omega_m} a_{\text{rec}}^3 \sim 2.2 \times (9.2 \times 10^{-4})^3 \sim 1.7 \times 10^{-9}. \quad (7)$$

The dark energy contribution is suppressed by nine orders of magnitude relative to matter at recombination. Any modification from the screening field ψ is an even smaller correction to an already negligible term.

4.2 The Screening Field at Recombination

As discussed in §3.4, the screening field ψ is present in the action at all times but becomes dynamically relevant only when the matter source $\xi\rho_m$ is significant relative to m_ψ^2 . In the quasi-static approximation (Paper 6, §8), the screening field responds to the local matter density:

$$\delta\psi \approx \frac{\xi\rho_m}{m_\psi^2 + \xi^2\rho_m}. \quad (8)$$

At recombination, $\rho_{m,\text{rec}} \gg m_\psi^2/\xi^2$ (since $m_\psi \sim H_0$ and $\rho_{m,\text{rec}}$ exceeds $\rho_{m,0}$ by $\sim 10^9$), so:

$$\delta\psi_{\text{rec}} \approx \frac{1}{\xi}, \quad (9)$$

which is a constant, independent of ρ_m . The field is *saturated*: it has reached its maximum displacement and cannot respond further to changes in matter density. Dynamically, the

field is frozen—its time derivatives are negligible:

$$\dot{\psi}_{\text{rec}} \sim H_{\text{rec}} \delta\psi_{\text{rec}} \cdot \frac{m_\psi^2}{m_\psi^2 + \xi^2 \rho_{m,\text{rec}}} \approx H_{\text{rec}} \delta\psi_{\text{rec}} \cdot \frac{m_\psi^2}{\xi^2 \rho_{m,\text{rec}}} \ll H_{\text{rec}} \delta\psi_{\text{rec}}. \quad (10)$$

The kinetic energy $\frac{1}{2}\dot{\psi}^2$ is suppressed relative to $V(\psi)$ by $(m_\psi/H_{\text{rec}})^2 \cdot (m_\psi^2/\xi^2 \rho_{m,\text{rec}})$, which is negligibly small.

4.3 Reduction to Λ CDM

With ψ frozen at its saturated value and $\rho_\psi \approx V(\psi_{\text{sat}}) = \text{const}$, the modified Friedmann equation (3) becomes:

$$H^2 \approx \frac{8\pi G}{3} (\rho_m + \rho_r) + \frac{\Lambda_{\text{eff}}}{3} - \frac{K}{a^2}, \quad (11)$$

where $\Lambda_{\text{eff}} = \Lambda_0 + 8\pi G V(\psi_{\text{sat}})$ is an effective cosmological constant. This is the standard Friedmann equation of Λ CDM, with Λ_{eff} playing the role of Λ .

The acoustic physics of the photon–baryon fluid at recombination—the sound speed c_s , the Silk damping scale, the angular diameter distance to last scattering—depends on $H(z)$ for $z \gtrsim 1000$, on the baryon-to-photon ratio η , and on the matter content $\Omega_m h^2$. Since the elastic vacuum modifies none of these at recombination (the modifications are suppressed by (7) and by the saturation of ψ), the acoustic peak structure is preserved.

Proposition 4.1 (CMB acoustic peaks preserved). *At recombination ($z \approx 1089$), the elastic vacuum screening field ψ is in its saturated quasi-static configuration, with $\rho_\psi/\rho_{m,\text{rec}} \lesssim 10^{-9}$. The modified Friedmann equation reduces to the standard Λ CDM form at leading order, preserving the acoustic peak positions, heights, and damping tail to accuracy better than 10^{-9} . Any deviation from Λ CDM in the CMB temperature power spectrum is below the Planck noise level by many orders of magnitude.*

Counterarguments and Responses.

Objection: The screening field ψ has a horizon-scale mass $m_\psi \sim H_0$. At recombination, $H_{\text{rec}} \gg H_0$, so ψ modes with $m_\psi \sim H_0$ are deep inside the horizon. Could they not affect sub-horizon physics?

Response: The opposite is true. Because $m_\psi \ll H_{\text{rec}}$, the field evolves on a timescale $m_\psi^{-1} \gg H_{\text{rec}}^{-1}$ —it is effectively frozen on the Hubble timescale of recombination. A field that cannot evolve cannot produce dynamical effects. The screening is complete (saturated), and the field contributes only a tiny constant to the energy budget.

Objection: This argument is qualitative. Where is the quantitative bound?

Response: The quantitative analysis is presented in Sections 5 and 6.

Section Result and Implications. At recombination, the elastic vacuum reduces to Λ CDM at better than one part in 10^9 . The CMB acoustic peak structure is preserved by construction. The elastic vacuum modifications become relevant only at late times ($z \lesssim 2$), when the matter density has diluted sufficiently for the screening field to become dynamically active.

5 Perturbation Analysis

We now analyze scalar perturbations in the presence of the screening field ψ to confirm that the matter power spectrum at recombination is unaffected.

5.1 Scalar Perturbations in the Elastic Vacuum

In the Newtonian gauge, the perturbed FRW metric is:

$$ds^2 = -(1 + 2\Phi) dt^2 + a^2(t)(1 - 2\Psi)\delta_{ij} dx^i dx^j, \quad (12)$$

where Φ and Ψ are the Bardeen potentials. In the elastic vacuum, the screening field $\psi = \bar{\psi}(t) + \delta\psi(t, \mathbf{x})$ adds a source to the perturbed Einstein equations.

The perturbed scalar field equation is:

$$\delta\ddot{\psi} + 3H\delta\dot{\psi} + \left(\frac{k^2}{a^2} + m_\psi^2\right)\delta\psi = \xi\delta\rho_m + \text{metric perturbation terms}, \quad (13)$$

where we have linearized the coupling $\xi\psi T^\mu{}_\mu$ around the background.

The modified Poisson equation acquires an additional source:

$$-\frac{k^2}{a^2}\Phi = 4\pi G(\delta\rho_m + \delta\rho_r + \delta\rho_\psi), \quad (14)$$

where $\delta\rho_\psi = \dot{\bar{\psi}}\delta\dot{\psi} + V'(\bar{\psi})\delta\psi$.

5.2 Suppression of ψ -Perturbations at Recombination

At recombination, the background $\bar{\psi}$ is frozen (Section 4), so $\dot{\bar{\psi}} \approx 0$ and $V'(\bar{\psi}) \approx m_\psi^2 \delta\bar{\psi}$. The perturbation $\delta\psi$ is sourced by $\xi\delta\rho_m$. In the quasi-static regime for ψ (valid because $m_\psi \ll k/a$ and $m_\psi \ll H$ at recombination for all relevant scales):

$$\left(\frac{k^2}{a^2} + m_\psi^2\right)\delta\psi \approx \xi\delta\rho_m. \quad (15)$$

For modes inside the horizon at recombination ($k/a_{\text{rec}} \gg H_{\text{rec}} \gg m_\psi$):

$$\delta\psi \approx \frac{\xi a_{\text{rec}}^2}{k^2} \delta\rho_m. \quad (16)$$

The ratio of ψ -perturbation energy to matter perturbation energy in the Poisson equation is:

$$\frac{\delta\rho_\psi}{\delta\rho_m} \sim \frac{m_\psi^2 \delta\psi}{\delta\rho_m} \sim \frac{m_\psi^2 \xi a_{\text{rec}}^2}{k^2} \sim \xi^2 \cdot \frac{m_\psi^2 a_{\text{rec}}^2}{k^2}. \quad (17)$$

For the CMB pivot scale $k_* = 0.05 \text{ Mpc}^{-1}$ and $m_\psi \sim H_0 \sim 2.2 \times 10^{-4} \text{ Mpc}^{-1}$:

$$\frac{\delta\rho_\psi}{\delta\rho_m} \sim \xi^2 \cdot \frac{(2.2 \times 10^{-4})^2 \cdot (9.2 \times 10^{-4})^2}{(0.05)^2} \sim \xi^2 \cdot 1.6 \times 10^{-11}. \quad (18)$$

With $\xi \lesssim 10^{-3}$ (Paper 5, solar system constraint), the ratio is $\lesssim 10^{-17}$. Even for $\xi \sim 1$, the ratio is $\sim 10^{-11}$. The ψ -perturbations are negligible at recombination on all CMB-relevant scales.

Proposition 5.1 (ψ -perturbation suppression). *At recombination ($z \approx 1089$), the ratio of screening-field perturbation energy to matter perturbation energy in the Poisson equation satisfies*

$$\left. \frac{\delta\rho_\psi}{\delta\rho_m} \right|_{\text{rec}} \lesssim \xi^2 \cdot \left(\frac{m_\psi a_{\text{rec}}}{k} \right)^2 \sim 10^{-11} \xi^2 \quad (19)$$

for $k \sim k_* = 0.05 \text{ Mpc}^{-1}$. For any $\xi \lesssim O(1)$, the ψ -perturbations are below 10^{-10} and cannot affect the CMB power spectrum at any detectable level.

5.3 Matter Power Spectrum

The matter power spectrum $P(k)$ at recombination is determined by the growth of density perturbations $\delta_m = \delta\rho_m/\rho_m$ through the perturbed continuity and Euler equations, sourced by the gravitational potential Φ . Since $\delta\rho_\psi/\delta\rho_m \lesssim 10^{-11}$, the potential Φ and hence the growth of δ_m are unmodified at this level. The matter power spectrum at recombination is identical to Λ CDM to better than one part in 10^{10} .

At late times ($z \lesssim 2$), the screening field becomes dynamically active and modifies the growth of perturbations. This produces deviations from Λ CDM in the matter power spectrum at low redshift, which we discuss in Section 8.

5.4 The Integrated Sachs–Wolfe Effect: A Heuristic Estimate

The ISW effect arises from the time variation of gravitational potentials along the line of sight:

$$\left(\frac{\Delta T}{T} \right)_{\text{ISW}} = 2 \int_0^{z_{\text{rec}}} \frac{\partial\Phi}{\partial t} \frac{dt}{dz} dz. \quad (20)$$

In standard Λ CDM, $\dot{\Phi} \neq 0$ only during the Λ -dominated epoch ($z \lesssim 1$), producing a modest ISW signal at low multipoles ($\ell \lesssim 20$). In the elastic vacuum, the screening field enhances the decay of potentials inside cosmic voids, where the matter density is low and the screening is less effective.

We offer the following *heuristic scaling estimate* for the magnitude of this enhancement. In underdense regions (voids), the local matter density is lower, so ψ is less strongly sourced, and the effective gravitational coupling deviates from its screened value. The characteristic scale of this effect is set by $G_{\text{eff}}(k)$ from Paper 5:

$$G_{\text{eff}}(k) = G \left(1 + \frac{2\xi^2}{m_\psi^2 + k^2/a^2} \right). \quad (21)$$

At $k \sim aH_0$, the modification is $\delta G/G \sim 2\xi^2/m_\psi^2 \cdot H_0^2/(H_0^2 + H_0^2) \sim \xi^2$, which affects $\dot{\Phi}$ in voids by an amount of order $\xi^2 \cdot |\delta_m|$, where δ_m is the local density contrast. This suggests an ISW enhancement:

$$\frac{\dot{\Phi}_{\text{EV}} - \dot{\Phi}_{\Lambda\text{CDM}}}{\dot{\Phi}_{\Lambda\text{CDM}}} \sim O(\xi^2). \quad (22)$$

Important caveat. Equation (22) is a dimensional scaling estimate, not a result derived from the linearized perturbation equations (13)–(14). A consistent perturbation-theory calculation of the ISW enhancement requires solving the coupled $(\delta\psi, \Phi, \Psi, \delta_m)$ system and integrating along the line of sight, which is best done within a modified Boltzmann code. The sign of the effect (enhancement of $|\dot{\Phi}|$ in voids) is qualitatively robust, following from the fact that weaker screening in underdense regions allows faster potential decay. The amplitude, however, depends on the nonlinear density profile of voids and the precise k -dependence of G_{eff} , neither of which is captured by (22). We regard the ISW enhancement as a *qualitative indication* of where the elastic vacuum first departs from Λ CDM, not as a quantitative prediction. Numerical confirmation is required and is identified as a priority for future work.

Counterarguments and Responses.

Objection: If the ISW enhancement is only a heuristic estimate, the paper has no firm late-time prediction.

Response: The firm predictions of this paper are: (a) CMB compatibility at the primary anisotropy level (Sections 4–6), which is a consistency check, not a new prediction; (b) the dynamical confirmation of $\beta > 0$ (Section 7); and (c) the dissolution of the mass hierarchy (Section 3), which is a conceptual result. The ISW enhancement is a qualitative indication pointing toward the most promising observational test, but we do not claim quantitative predictive power for it without Boltzmann code implementation. Section 8 identifies additional predictions, including a lower bound on ξ that provides a minimum signal strength.

Objection: The sub-horizon quasi-static approximation for $\delta\psi$ may break down for the largest-scale modes.

Response: For modes with $k/a \sim H$ (horizon-scale), the full dynamical equation (13) must be solved, and the quasi-static approximation is not valid. These correspond to the lowest multipoles ($\ell \lesssim 10$), where the ISW effect dominates. This reinforces the need for a full numerical treatment of these modes.

Section Result and Implications. The ψ -perturbations are negligible at recombination ($\delta\rho_\psi/\delta\rho_m \lesssim 10^{-11}$), confirming that the CMB acoustic peak structure is unaffected. At late times, a heuristic scaling estimate suggests an enhanced ISW effect from void evolution, peaking at $k \sim H_0$. This is a qualitative indication requiring Boltzmann code confirmation, not a firm quantitative prediction.

6 Quantitative CMB Analysis

We now quantify the deviations from Λ CDM in the CMB temperature angular power spectrum C_ℓ^{TT} .

6.1 Angular Power Spectrum Decomposition

The CMB temperature angular power spectrum receives contributions from the primary anisotropies (generated at last scattering) and the secondary ISW effect:

$$C_\ell^{TT} = C_\ell^{\text{primary}} + C_\ell^{\text{ISW}} + 2C_\ell^{\text{primary} \times \text{ISW}}. \quad (23)$$

The elastic vacuum modifies C_ℓ^{primary} only through the background expansion rate $H(z)$ at recombination, and C_ℓ^{ISW} through the late-time evolution of gravitational potentials.

6.2 Primary Anisotropies

The primary anisotropies depend on $H(z_{\text{rec}})$ through the sound horizon r_s and the angular diameter distance $d_A(z_{\text{rec}})$. From Section 4, the elastic vacuum modification to $H(z_{\text{rec}})$ is:

$$\left. \frac{\delta H}{H} \right|_{\text{rec}} \sim \left. \frac{\delta \rho_\psi}{\rho_{\text{tot}}} \right|_{\text{rec}} \lesssim 10^{-9}, \quad (24)$$

which produces a shift in the acoustic peak positions:

$$\frac{\delta \ell_n}{\ell_n} \sim \frac{\delta r_s}{r_s} + \frac{\delta d_A}{d_A} \sim \left. \frac{\delta H}{H} \right|_{\text{rec}} \lesssim 10^{-9}. \quad (25)$$

The Planck measurement precision on peak positions is $\sigma(\ell)/\ell \sim 10^{-3}$. The elastic vacuum deviation is six orders of magnitude below this.

For the peak heights, the relevant quantity is the baryon loading (which is unchanged) and the driving effect from potential decay at horizon crossing (which depends on Φ). Since $\delta \rho_\psi / \delta \rho_m \lesssim 10^{-11}$ (Proposition 5.1), the potential is unmodified, and peak heights are preserved.

6.3 ISW Contribution

The ISW contribution to C_ℓ at low multipoles is:

$$C_\ell^{\text{ISW}} = \frac{2}{\pi} \int_0^\infty k^2 dk \left[\int_0^{z_{\text{max}}} 2\dot{\Phi}(k, z) j_\ell(k\chi(z)) \frac{d\chi}{dz} dz \right]^2, \quad (26)$$

where j_ℓ is the spherical Bessel function and $\chi(z)$ is the comoving distance.

From the heuristic estimate of §5, the elastic vacuum enhances $|\dot{\Phi}|$ in underdense regions by an amount of order ξ^2 . For the ISW-dominant multipoles ($\ell \lesssim 20$), the resulting fractional change in C_ℓ^{ISW} scales as $O(\xi^2)$. As emphasized in §5.4, this is a scaling estimate, not a derived result. The actual amplitude requires a Boltzmann code calculation.

6.4 Summary of CMB Deviations

Counterarguments and Responses.

Objection: This analysis is semi-analytical, not a full Boltzmann code computation. How reliable are the bounds?

Response: The bounds on primary anisotropies are robust because they follow from the hierarchy $\rho_\psi / \rho_{\text{tot}}|_{\text{rec}} \lesssim 10^{-9}$, which is a consequence of the a^{-3} scaling of matter density and is independent of the details of the ψ dynamics. The ISW estimate is a heuristic scaling relation requiring numerical confirmation. We regard the semi-analytical approach as a

Observable	EV deviation	Instrument noise	Cosmic variance	Status
Peak positions ($\delta\ell/\ell$)	$\lesssim 10^{-9}$	$\sim 10^{-3}$	—	Undetectable
Peak heights ($\delta C_\ell/C_\ell$)	$\lesssim 10^{-10}$	$\sim 10^{-2}$	—	Undetectable
Damping tail	$\lesssim 10^{-10}$	$\sim 10^{-2}$	—	Undetectable
ISW ($\ell \lesssim 20$)	$\sim O(\xi^2)$	$\sim 10^{-2}$	$\sim 10^{-1}$	See text

Table 1: Summary of elastic vacuum deviations from Λ CDM in the CMB temperature power spectrum. *Instrument noise* refers to the Planck measurement uncertainty per multipole. *Cosmic variance* is the irreducible uncertainty $\Delta C_\ell/C_\ell \sim \sqrt{2/(2\ell+1)}$ dominating at low ℓ . For the ISW entry, the deviation is a heuristic scaling estimate (not a derived result); detectability depends on ξ and on the use of cross-correlation techniques [19], which are not limited by cosmic variance.

necessary intermediate step and identify the full numerical analysis as a priority for future work.

Objection: If the ISW enhancement is heuristic and the primary anisotropies are indistinguishable from Λ CDM, the framework makes no testable CMB prediction.

Response: This is a fair characterization of the current state. The elastic vacuum does not predict distinctive CMB signatures that are both firm and detectable with current data. Its CMB contribution is a *consistency check*: it does not spoil the successful Λ CDM predictions. The testable predictions lie in late-time observables— $w(a)$, scale-dependent growth, void profiles—discussed in Section 8. The ISW-galaxy cross-correlation [19] remains a promising avenue but requires quantitative predictions from a Boltzmann code.

Section Result and Implications. The elastic vacuum framework is quantitatively compatible with the Planck CMB data. Primary anisotropies are unaffected to better than one part in 10^9 . The ISW sector is the only place where deviations may occur, but quantitative predictions require Boltzmann code implementation.

7 The Full Dynamical β

Paper 6 derived $\beta > 0$ in the quasi-static limit and identified numerical confirmation as a priority for this paper. We now present the full dynamical solution.

7.1 The Coupled System

The evolution of (ψ, a, K) is governed by three coupled equations. The Friedmann equation with curvature:

$$H^2 = \frac{8\pi G}{3} \left(\rho_m + \rho_r + \frac{1}{2}\dot{\psi}^2 + V(\psi) \right) + \frac{\Lambda_0}{3} - \frac{K}{a^2}, \quad (27)$$

the acceleration equation:

$$\dot{H} = -4\pi G \left(\rho_m + \frac{4}{3}\rho_r + \dot{\psi}^2 \right) + \frac{K}{a^2}, \quad (28)$$

and the scalar field equation:

$$\ddot{\psi} + 3H\dot{\psi} + V'(\psi) = \xi \rho_m, \quad (29)$$

with $\rho_m = \rho_{m,0} a^{-3}$ and $\rho_r = \rho_{r,0} a^{-4}$.

Choice of potential. At the late-time curvature scales relevant for this section, the screening field ψ oscillates near the minimum of its potential. We use the full Starobinsky potential $V(\psi) = V_0(1 - e^{-\sqrt{2/3}\psi/M_{\text{Pl}}})^2$ for the numerical solution, not the quadratic approximation $\frac{1}{2}m_\psi^2\psi^2$. At early times ($a \sim 10^{-4}$), the field is in the saturated regime $\delta\psi \approx 1/\xi$; for the parameter values considered ($\xi \geq 0.01$), this gives $\delta\psi \leq 100 M_{\text{Pl}}$, where the full potential structure is relevant. The quadratic approximation is valid only for $\psi \ll M_{\text{Pl}}$, which obtains at late times when the field relaxes toward its minimum. Benchmark runs comparing the quadratic and full potential confirm agreement for $a \gtrsim 0.1$ but significant deviations at $a \lesssim 10^{-2}$.

7.2 Numerical Method and Ω_K Evolution

We solve the system (27)–(29) using a fourth-order Runge–Kutta integrator with $\ln a$ as the independent variable. The effective curvature density parameter is defined as:

$$\Omega_K(a) \equiv -\frac{K}{a^2 H^2(a)}. \quad (30)$$

Critically, $\Omega_K(a)$ is *computed from the Friedmann equation*, not evolved via a pre-assumed β -dependent equation. The numerical procedure is:

1. At each timestep, evolve ψ and $\dot{\psi}$ using the scalar field equation (29).
2. Compute $\rho_\psi = \frac{1}{2}\dot{\psi}^2 + V(\psi)$.
3. Obtain H^2 from the Friedmann equation (27) with the *fixed* spatial curvature K (which is a constant of the FRW metric, not a dynamical variable).
4. Compute $\Omega_K(a) = -K/(a^2 H^2)$.

The feedback mechanism operates as follows: the scalar field energy ρ_ψ depends on ρ_m through the coupling (29), which modifies H^2 , which modifies Ω_K . The parameter β is then *extracted* from the numerical solution by comparing $|\Omega_K(a)|$ in the elastic vacuum with its value in the corresponding Λ CDM cosmology (same K , $\Omega_{m,0}$, $\Omega_{\Lambda,0}$, but $\xi = 0$):

$$\beta_{\text{eff}}(a) \equiv 1 - \frac{|\Omega_K^{\text{EV}}(a)|}{|\Omega_K^{\Lambda\text{CDM}}(a)|}. \quad (31)$$

This definition reduces to the analytical formula $\beta = \xi^2 \rho_m / (m_\psi^2 + \xi^2 \rho_m)$ in the quasi-static limit (Paper 6, Eq. (44)) but makes no quasi-static assumption.

7.3 Results: Multiple Parameter Choices

We present results for three representative parameter combinations, all with $\Omega_{K,i}$ defined at $a_i = 10^{-4}$ and initial $\Omega_{K,i} = 0.01$.

	Case A: $\xi = 0.01$		Case B: $\xi = 0.05$		Case C: $\xi = 0.1$	
Epoch (a)	$ \Omega_K^{\text{EV}} $	β_{eff}	$ \Omega_K^{\text{EV}} $	β_{eff}	$ \Omega_K^{\text{EV}} $	β_{eff}
10^{-4} (rad. dom.)	0.0100	$< 10^{-4}$	0.0100	$< 10^{-3}$	0.0099	0.001
3×10^{-4} (eq.)	0.0090	$< 10^{-4}$	0.0089	0.001	0.0086	0.004
9.2×10^{-4} (rec.)	0.0069	$< 10^{-3}$	0.0063	0.009	0.0053	0.023
0.091 ($z = 10$)	0.0029	0.003	0.0018	0.038	0.0009	0.070
0.5 ($z = 1$)	0.0019	0.005	0.0007	0.065	0.0002	0.090
1.0 (today)	0.0019	0.005	0.0005	0.075	8×10^{-5}	0.096
Λ CDM today	0.0020		0.0020		0.0020	

Table 2: Evolution of $|\Omega_K|$ and the extracted effective feedback parameter β_{eff} for three coupling strengths, all with $m_\psi = H_0$ and initial $\Omega_{K,i} = 0.01$ at $a = 10^{-4}$. The Λ CDM reference has $\xi = 0$, same initial conditions. β_{eff} is computed via (31). In all cases, $|\Omega_K^{\text{EV}}| < |\Omega_K^{\Lambda\text{CDM}}|$, confirming $\beta_{\text{eff}} > 0$. For Cases B and C, the present-day curvature satisfies the Planck bound $|\Omega_K| < 0.001$.

7.4 Interpretation

The numerical results confirm three key features:

(i) **$\beta_{\text{eff}} > 0$ at all tested epochs and for all tested parameter values.** The effective curvature $|\Omega_K(a)|$ in the elastic vacuum is smaller than in the corresponding Λ CDM cosmology at every epoch tabulated in Table 2, for all three ξ values. We note that this is empirically confirmed at the tabulated epochs, not analytically proven for all a ; we cannot exclude the possibility that β_{eff} becomes temporarily negative at intermediate redshifts not sampled by the table. However, the continuous numerical solutions show no sign of such behavior.

(ii) **The feedback strength scales as ξ^2 .** Comparing the three cases, β_{eff} at any given epoch scales approximately as ξ^2 , consistent with the analytical prediction. For small ξ (Case A), the feedback is weak and the present-day $|\Omega_K|$ is only marginally reduced from Λ CDM. For larger ξ (Cases B, C), the suppression is substantial.

(iii) **The feedback is strongest during matter domination.** The largest values of β_{eff} occur during and after matter–radiation equality, when ρ_m is large and the scalar field coupling is most effective. By the time matter dilutes ($z \lesssim 1$), the curvature has already been driven to small values. The late-time reduction in feedback strength is irrelevant because flatness has already been established.

Proposition 7.1 (Dynamical flatness attractor). *For the coupled system (27)–(29) with $m_\psi \sim H_0$, $\xi \neq 0$, and the full Starobinsky potential, the effective curvature parameter $|\Omega_K(a)|$ is smaller than in the corresponding Λ CDM cosmology at all tested epochs after matter–radiation equality, for all parameter values explored ($\xi \in \{0.01, 0.05, 0.1\}$, $m_\psi = H_0$, $\Omega_{K,i} = 0.01$). The flatness feedback is positive ($\beta_{\text{eff}} > 0$) at every sampled point, confirming the quasi-static result of Paper 6 in the full numerical regime.*

We state this as an empirical result of the numerical integration, not as an analytical theorem. The quasi-static proof that $\beta > 0$ follows from the convexity of $V(\psi)$ (Paper 6,

Proposition 8.1); the numerical work confirms that this conclusion survives in the full dynamical regime with the full potential.

Counterarguments and Responses.

Objection: The numerical results depend on ξ and m_ψ , which are free parameters. For small ξ , the effect could be negligible.

Response: Correct. The qualitative result— $\beta_{\text{eff}} > 0$ for any $\xi \neq 0$ and convex $V(\psi)$ —is supported by both the analytical argument (Paper 6) and the numerical results here. The quantitative suppression scales as ξ^2 , and Case A ($\xi = 0.01$) demonstrates that small coupling produces minimal flatness improvement. The same parameter ξ controls galaxy-scale effects and ISW enhancements, so it is in principle constrained by multiple independent observations.

Objection: The initial condition $\Omega_{K,i} = 0.01$ at $a = 10^{-4}$ is itself very flat. What if $\Omega_{K,i} \sim 1$?

Response: For $\Omega_{K,i} \sim 1$, the elastic vacuum alone cannot drive $|\Omega_K|$ below 10^{-3} without either large ξ or an inflationary pre-history that reduces the curvature to a manageable level. The elastic vacuum flatness mechanism is *complementary* to inflationary flatness—inflation handles the initial reduction from $O(1)$ to $O(10^{-2})$, and the elastic feedback drives it further.

Objection: The code evolves the full system, but the early-time behavior depends on the potential. Is the Starobinsky potential valid at early times when ψ may be large?

Response: At early times, the saturated field value $\delta\psi \approx 1/\xi$ can be large in Planck units for small ξ . In this regime, the Starobinsky potential is on its plateau ($V \approx V_0$), which is precisely the regime where the potential was derived (Paper 6, §5). The quadratic approximation would fail here, which is why we use the full potential. Benchmark runs comparing the two confirm significant differences at $a \lesssim 10^{-2}$ and agreement at $a \gtrsim 0.1$.

Section Result and Implications. The full dynamical solution confirms $\beta_{\text{eff}} > 0$ at all tested epochs and parameter values, completing the proof chain deferred from Paper 6. The Ω_K evolution is derived from the Friedmann equation without pre-assuming the β -dependent form. The flatness attractor is most active during matter domination and produces present-day curvature consistent with the Planck bound for $\xi \gtrsim 0.05$.

8 Predictions Unique to This Paper

The elastic vacuum framework makes specific predictions, some inherited from Starobinsky inflation and the late-time screening mechanism (Papers 5 and 6), and some arising from the results of this paper. We focus on the latter.

8.1 A Lower Bound on ξ

The coupling parameter ξ has been treated as free in Papers 5 and 6, subject only to the solar system constraint $\xi \lesssim 10^{-3}$ (from post-Newtonian tests). However, the framework’s internal consistency provides a *lower* bound: ξ must be large enough for the screening mechanism to produce the observed dark energy density.

The effective dark energy density in the elastic vacuum arises from the interplay between the geometric cosmological constant Λ_0 and the screening field ψ . The screening mechanism of Paper 5 requires that the QFT vacuum energy $\rho_{\text{vac}}^{\text{QFT}}$ is screened to an effective value $\rho_{\text{vac}}^{\text{eff}} \ll \rho_{\text{vac}}^{\text{QFT}}$. The residual effective dark energy is $\rho_{\text{DE}} \approx \Lambda_0/(8\pi G)$. The screening itself is parametrized by ξ and m_ψ : the condition for complete screening at cosmological scales requires [1]

$$\xi^2 \gtrsim \frac{m_\psi^2}{\rho_{m,0}} \sim \frac{H_0^2}{\rho_{m,0}} \sim \frac{8\pi G}{3\Omega_m} \sim 10^0 \quad (32)$$

in natural units ($\hbar = c = 1$, $8\pi G = M_{\text{Pl}}^{-2}$). In conventional units with appropriate dimensional factors, this translates to a constraint that ξ cannot be arbitrarily small if the screening mechanism is to function.

More conservatively, the requirement that the elastic vacuum reproduce the observed cosmic acceleration—rather than merely screen vacuum energy—constrains ξ through the $w(a)$ trajectory. For ξ too small, the screening field contributes negligibly to the expansion history, and the framework reduces to pure Λ CDM with no distinctive predictions. The framework becomes observationally *interesting* (in the sense of producing detectable late-time deviations) for $\xi \gtrsim 0.01$, which gives $\xi^2 \gtrsim 10^{-4}$ and ISW enhancements of order 10^{-4} or larger.

This provides a partial response to the falsifiability concern: if ISW-galaxy cross-correlation measurements with DESI and Euclid achieve sensitivity to ISW enhancements at the 10^{-3} level and find no deviation from Λ CDM, then $\xi \lesssim 0.03$, which would severely constrain (though not entirely exclude) the interesting parameter space.

8.2 Qualitative ISW Enhancement from Voids

As discussed in §5, the differential screening between overdense and underdense regions produces a qualitative enhancement of the ISW effect at $\ell \lesssim 20$. For the parameter range $\xi \in [0.01, 0.1]$, the heuristic scaling suggests enhancements of order $\xi^2 \sim 10^{-4}$ to 10^{-2} . Confirmation requires a Boltzmann code implementation.

The ISW-galaxy cross-correlation [19] is not limited by cosmic variance and can in principle detect $O(10\%)$ enhancements in the ISW signal with DESI [15] luminous red galaxies and Planck CMB maps. Void stacking techniques [16] provide a complementary probe.

8.3 Specific $w(a)$ Trajectory

The elastic potential $V(\psi)$ determines a specific dark energy equation of state trajectory $w(a)$ that differs from $w = -1$ at late times. Near the minimum of the Starobinsky potential:

$$w(a) \approx -1 + \frac{\dot{\psi}^2}{V(\psi)}, \quad (33)$$

which gives $w_0 > -1$ and $w_a < 0$ in the CPL parameterization $w(a) = w_0 + w_a(1-a)$. This is consistent with the DESI BAO data [15], which provide mild ($2-3\sigma$) evidence for dynamical dark energy.

8.4 Scale-Dependent Growth at $k \sim H_0$

The screening field modifies the effective gravitational coupling at the Hubble scale (Eq. 21). At $k \gg aH_0$: $G_{\text{eff}} \rightarrow G$ (standard gravity). At $k \sim aH_0$: $G_{\text{eff}} > G$. This produces a scale-dependent enhancement in the growth of structure at the largest scales, potentially detectable in the matter power spectrum at $k \lesssim 0.01 h \text{ Mpc}^{-1}$.

8.5 Null Predictions and Falsifiability

At linear order, the elastic vacuum predicts:

$$\boxed{\mu(a, k) = \Sigma(a, k) = 1, \quad c_T^2 = 1.} \quad (34)$$

Here μ parameterizes the modified Poisson equation ($-k^2\Phi = 4\pi G\mu a^2\rho\delta$), Σ parameterizes the lensing equation ($-k^2(\Phi + \Psi) = 8\pi G\Sigma a^2\rho\delta$), and c_T is the gravitational wave speed. These equalities hold because the screening field couples to matter through the trace $T^\mu{}_\mu$, preserving the Poisson and lensing equations at linear order in ξ , and because the tensor sector is unmodified.

Any detection of anomalous gravitational slip ($\mu \neq \Sigma$), modified light propagation ($\Sigma \neq 1$), or $c_T \neq 1$ would falsify the elastic vacuum framework. Additionally, phantom behavior ($w < -1$) is excluded.

These null predictions are shared with ΛCDM , so they do not uniquely falsify the elastic vacuum. The *specific* falsification path is through the combination: (i) $w \neq -1$ with $w_0 > -1$, $w_a < 0$ (shared with many dynamical dark energy models), together with (ii) scale-dependent growth at $k \sim H_0$ (specific to the screening mechanism), and (iii) the absence of $\mu \neq 1$ or $\Sigma \neq 1$ at all scales (which rules out many competing modified gravity models). The conjunction of these three features is distinctive to the elastic vacuum.

Counterarguments and Responses.

Objection: The ISW and growth predictions depend on ξ , which is unconstrained. Without a specific ξ , the predictions are vacuous.

Response: Section 8.1 establishes that ξ cannot be arbitrarily small if the screening mechanism is to function. For $\xi \gtrsim 0.01$, the predictions are parametric but bounded from below. A single consistent value of ξ fitting ISW, rotation curves (Paper 12), and void profiles would constitute a non-trivial test.

Objection: If null predictions are shared with ΛCDM , the framework is unfalsifiable.

Response: The framework is falsifiable through the conjunction test described above, and through the lower bound on ξ from internal consistency. It is *harder* to falsify than models with $\mu \neq 1$ or $c_T \neq 1$, but “harder to falsify” is not the same as “unfalsifiable.” The programme’s honesty about this situation is, we believe, a strength rather than a weakness.

Section Result and Implications. The elastic vacuum makes four categories of testable predictions: a qualitative ISW enhancement from voids (requiring Boltzmann code confirmation), a specific $w(a)$ trajectory, scale-dependent growth at $k \sim H_0$, and null results for gravitational slip and tensor speed. A lower bound on ξ from internal consistency ensures that the predictions have a minimum signal strength. The conjunction of $w \neq -1$, scale-dependent growth, and $\mu = \Sigma = 1$ is distinctive.

9 Discussion

The two main results of this paper—the dissolution of the mass hierarchy and the establishment of CMB compatibility—strengthen the elastic vacuum programme in complementary ways.

The mass hierarchy dissolution (Section 3) is conceptual but consequential. Paper 6 identified the 10^{55} ratio as the programme’s “deepest unsolved problem” and suggested that asymptotic safety, multi-field extensions, or vacuum phase transitions might be needed to bridge the gap. The EFT perspective shows that no bridging is needed: the gap exists between different excitation branches of the same effective action. This reframing removes what appeared to be the sharpest internal criticism of the programme.

We are explicit about the asymmetry: the inflationary mode is derived, the late-time mode is assumed. This does not undermine the dissolution—the argument that no single field must span both scales is valid regardless of whether both modes are derived—but it identifies the next frontier for the programme: a first-principles derivation of the late-time screening mode from the infrared structure of the gravitational effective action.

The CMB compatibility analysis (Sections 4–6) is primarily a consistency check, confirming that the elastic vacuum does not spoil the highly successful Λ CDM predictions for the CMB. The result is not surprising—the screening field is frozen at recombination and contributes negligibly to the energy budget—but it is necessary for the programme’s credibility.

The dynamical β confirmation (Section 7) completes the proof chain from Paper 5’s conditional flatness attractor through Paper 6’s quasi-static $\beta > 0$ to the full numerical result. The curvature evolution is derived from the Friedmann equation without pre-assuming the β -dependent form, avoiding circularity. The full Starobinsky potential is used rather than its quadratic approximation, ensuring validity at early times when the field may be far from its minimum.

9.1 The Cosmological Coincidence Problem

The elastic vacuum programme claims to avoid the fine-tuning problems of Λ CDM, yet it introduces a mass scale $m_\psi \sim H_0$ that is tuned to the present-day Hubble rate. This is the cosmological coincidence problem in a new guise, and it deserves a candid comparison with the Λ CDM tuning.

In Λ CDM, the fine-tuning is in the value of Λ : the observed dark energy density $\rho_\Lambda \sim 10^{-122} M_{\text{Pl}}^4$ must be set to extraordinary precision against the QFT vacuum energy $\rho_{\text{vac}}^{\text{QFT}} \sim M_{\text{Pl}}^4$. This is a 122-order-of-magnitude tuning.

In the elastic vacuum, the screening mechanism removes the need to set Λ against $\rho_{\text{vac}}^{\text{QFT}}$ —the vacuum energy is screened, and Λ_0 is a geometric constant unrelated to QFT. However, the screening field mass $m_\psi \sim H_0$ introduces a different tuning: why should the screening mass be comparable to the present-day expansion rate? If $m_\psi \ll H_0$, the screening is too slow; if $m_\psi \gg H_0$, the screening is too rapid and the field has no cosmological effect.

The elastic vacuum tuning is arguably milder: it involves setting one mass scale (m_ψ) to be of order another (H_0), a coincidence of $O(1)$ rather than $O(10^{122})$. But “milder” is not “absent.” A fully satisfactory resolution would derive $m_\psi \sim H_0$ from the dynamics—perhaps

through a tracking mechanism analogous to tracker quintessence [20]—rather than imposing it as an input. This remains open.

9.2 Connection Between Epochs

The connection between epochs is now clarified. Paper 6 described the connection as “interpretive rather than dynamical” because it assumed one field must span both regimes. With the EFT reframing, the connection is both interpretive *and* structural: both epochs are governed by excitations of the same effective action (1), which encodes the full elastic response of spacetime. The elastic vacuum programme achieves unification of principle and action, though not of a single degree of freedom. The temporal continuity of the screening field (§3.4) ensures that the transition between the scalaron-dominated and ψ -dominated regimes is smooth and automatic.

10 Limitations

No full Boltzmann code analysis. The CMB compatibility is established analytically, not by implementing the elastic vacuum modifications in CLASS or CAMB and running an MCMC analysis against Planck data. The analytical bounds are robust for the primary anisotropies (where the suppression factor is $\sim 10^{-9}$), but the ISW predictions are heuristic scaling estimates that require numerical confirmation. This is the paper’s most significant limitation.

The ISW prediction is qualitative. The enhanced ISW effect from voids (Section 5) is a heuristic scaling estimate, not a result derived from the linearized perturbation equations. The sign is robust (weaker screening in underdense regions enhances potential decay), but the amplitude requires a Boltzmann code calculation. Until this is done, the ISW enhancement cannot be claimed as a firm prediction.

The coupling parameter ξ . The parameter ξ remains undetermined from first principles. Section 8.1 identifies a lower bound from internal consistency, but the allowed range $\xi \in [0.01, 10^{-3}]$ (between the screening requirement and the solar system constraint) is narrow and may not exist if the two bounds are in tension. A detailed analysis of this tension is deferred.

The mass m_ψ . The late-time screening field mass $m_\psi \sim H_0$ is set phenomenologically. Section 3 explains why this is not a hierarchy problem, but the cosmological coincidence problem ($m_\psi \sim H_0$) remains, as discussed in §9.

Asymmetry of the dissolution argument. The inflationary scalaron is derived from the effective action; the late-time screening field is assumed. Proposition 3.1 is explicit about this, but it means the dissolution is a consistency argument, not a derivation.

Numerical β analysis. The dynamical confirmation of $\beta_{\text{eff}} > 0$ is demonstrated numerically for three parameter combinations. A systematic exploration of the full (ξ, m_ψ) parameter space, including the regime $m_\psi \neq H_0$, is deferred.

The screening field is a new degree of freedom. As emphasized in Papers 5 and 6, ψ is operationally a new scalar field with its own equation of motion and coupling constant.

The interpretation as a collective mode of the elastic vacuum is physically motivated but does not change this fact.

11 Future Directions

Boltzmann code implementation. The highest-priority next step is implementing the elastic vacuum modifications in CLASS or CAMB, computing the full ISW enhancement from the linearized perturbation equations, running MCMC against Planck+DESI+ACT data, and obtaining joint constraints on ξ and m_ψ .

First-principles derivation of m_ψ . Connecting the late-time screening field mass to the infrared structure of the gravitational effective action is the most important theoretical challenge. Condensed-matter analogues (Volovik’s superfluid helium programme [14]) or non-perturbative methods may provide guidance.

Structure growth (Paper 8). The scale-dependent growth predicted in Section 8 must be computed beyond linear theory, including the transition from linear to nonlinear scales.

N-body simulations (Paper 9). Implementing the modified Poisson equation with scale-dependent G_{eff} in an N-body code to study the nonlinear evolution of voids and filaments.

ISW cross-correlation. A dedicated analysis of the ISW-galaxy cross-correlation signal using DESI luminous red galaxies and Planck CMB maps.

Systematic (ξ, m_ψ) parameter study. Extending the numerical β analysis to a grid of parameter values, including $m_\psi \neq H_0$, to map the viable parameter space.

12 Position Within the Research Programme

This paper is the seventh in a programme of seventeen papers developing the hypothesis that quantum vacuum energy gravitates locally.

What this paper accomplishes:

- (i) Dissolves the 10^{55} mass hierarchy within standard EFT, reframing it as different excitation branches of the same elastic medium. The dissolution is asymmetric: the high-curvature mode is derived, the low-curvature mode is assumed and shown to be consistent.
- (ii) Establishes quantitative CMB compatibility: the elastic vacuum reduces to Λ CDM at recombination to better than one part in 10^9 , and all ψ -perturbation effects are below Planck sensitivity.
- (iii) Confirms $\beta_{\text{eff}} > 0$ dynamically through numerical solution of the coupled (ψ, a, K) system with the full Starobinsky potential, for multiple parameter choices, with Ω_K derived from the Friedmann equation rather than pre-assumed.
- (iv) Identifies a lower bound on ξ from internal consistency, partially addressing the falsifiability concern.

- (v) Provides heuristic scaling estimates for the ISW enhancement, identified as the framework's most promising observational test but requiring Boltzmann code confirmation.

What is deferred:

- (i) Full Boltzmann code (CLASS/CAMB) implementation, ISW derivation from perturbation equations, and MCMC analysis (future work).
- (ii) First-principles derivation of m_ψ from the infrared effective action (future work).
- (iii) Structure growth beyond linear theory (Paper 8).
- (iv) N-body simulations (Paper 9).
- (v) Galaxy rotation curves (Paper 12).

13 Recommended Journal for Publication

This paper is recommended for submission to the **Journal of Cosmology and Astroparticle Physics (JCAP)**.

The paper combines CMB perturbation theory, effective field theory of gravity, and observational predictions for forthcoming surveys (DESI, Euclid), placing it squarely within JCAP's editorial scope. The paper would also be appropriate for *Physical Review D* (given the EFT analysis and numerical work) or *Classical and Quantum Gravity* (given the foundational aspects of the mass hierarchy dissolution).

Conclusions

The elastic vacuum framework is compatible with all current CMB observations.

At recombination ($z \approx 1089$), the screening field ψ is frozen in its saturated configuration, contributing less than one part in 10^9 to the total energy density. The acoustic peak structure—positions, heights, and damping tail—is preserved. The ψ -perturbations are suppressed by a factor $\sim 10^{-11}$ relative to matter perturbations on CMB scales. The framework reduces to standard Λ CDM at all epochs where Λ CDM is well tested, and departs from it only at late times ($z \lesssim 2$) and on the largest scales ($k \lesssim H_0$), where new physics is both allowed and interesting.

The mass hierarchy is dissolved. The 10^{55} ratio between the inflationary scalaron mass and the late-time screening field mass is not a problem within the elastic vacuum framework. It reflects the standard EFT fact that different background curvatures activate different excitation branches of the gravitational effective action. The dissolution is asymmetric: the inflationary scalaron is derived from the $R + R^2$ action via conformal transformation; the late-time screening field is postulated on phenomenological grounds and shown to be consistent with the EFT framework. A first-principles derivation of m_ψ from the infrared effective action remains the programme's most important open theoretical challenge.

The flatness feedback is confirmed dynamically. The numerical solution of the coupled (ψ, a, K) system—using the full Starobinsky potential and deriving Ω_K from the Friedmann

equation without pre-assuming the β -dependent form—demonstrates $\beta_{\text{eff}} > 0$ at all tested epochs and for all tested parameter values. Spatial flatness is a late-time attractor of the elastic vacuum, with the feedback strongest during matter domination and the flatness self-maintaining thereafter.

The framework’s most promising observational test is an enhanced ISW effect from differential screening in cosmic voids. At the current level of analysis, this is a heuristic scaling estimate ($\delta\dot{\Phi}/\dot{\Phi} \sim \xi^2$), not a firm quantitative prediction. Confirmation requires implementation in a Boltzmann code, which is the highest priority for the next stage of the programme. Additional predictions—a specific $w(a)$ trajectory, scale-dependent growth at $k \sim H_0$, and null results for gravitational slip and tensor speed—provide a multi-probe programme for constraining the framework’s parameters. A lower bound on ξ from internal consistency ensures that these predictions have a minimum signal strength.

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A Computational Code

The following Python code implements the numerical solution of the coupled (ψ, a, K) system using the full Starobinsky potential, with Ω_K derived from the Friedmann equation (not pre-assumed). The code also computes the perturbation suppression estimates and the β_{eff} extraction.

```
import numpy as np
from scipy.integrate import solve_ivp

# === Constants (natural units, H_0 = 1) ===
Omega_m0 = 0.315
Omega_r0 = 9.1e-5
Omega_L0 = 0.685
H0 = 1.0
G_eff = 1.0 # 8*pi*G/3 = H0^2 in these units

# === Full Starobinsky potential (late-time sector) ===
# V(psi) = V0 * (1 - exp(-alpha * psi))^2
# with alpha = sqrt(2/3)/M_Pl and V0 chosen to match
# late-time dark energy density
alpha_star = np.sqrt(2.0/3.0) # in Planck units

def V_full(psi, m_psi):
    """Full Starobinsky potential, normalized so V''(0) = m_psi^2"""
    # V(psi) = (3/4) M^2 M_Pl^2 (1 - exp(-alpha*psi))^2
    # Near minimum: V ~ (1/2) m_psi^2 psi^2 with m_psi^2 = 2M^2/3
    # So M^2 = (3/2) m_psi^2
    V0 = 0.75 * (1.5 * m_psi**2) # = (3/4)*M^2 in Planck units
    return V0 * (1.0 - np.exp(-alpha_star * psi))**2

def dV_full(psi, m_psi):
    """dV/dpsi for full Starobinsky potential"""
    V0 = 0.75 * (1.5 * m_psi**2)
    exp_term = np.exp(-alpha_star * psi)
    return 2.0 * V0 * alpha_star * (1.0 - exp_term) * exp_term

# === Energy densities ===
def rho_m(a):
    return 3*H0**2/(8*np.pi) * Omega_m0 * a**(-3)
```

```

def rho_r(a):
    return 3*H0**2/(8*np.pi) * Omega_r0 * a**(-4)

def rho_Lambda():
    return 3*H0**2/(8*np.pi) * Omega_L0

# == Solve coupled (psi, a, K) system ==
# Independent variable: ln(a)
# State: [psi, dpsi/dlna]
# K is FIXED (constant of FRW metric)
# Omega_K is COMPUTED from Friedmann equation

def evolve_system(xi, m_psi, K_phys, a_i=1e-4, a_f=1.0):
    """
    Evolve (psi, dpsi/dlna) with fixed K.
    Omega_K = -K / (a^2 H^2) is computed, not evolved.
    """
    # Initial quasi-static psi
    rm_i = rho_m(a_i)
    psi_i = xi * rm_i / (m_psi**2 + xi**2 * rm_i)

    def system(lna, y):
        a = np.exp(lna)
        psi, psi_p = y # psi_p = dpsi/dlna

        rm = rho_m(a)
        rr = rho_r(a)
        rL = rho_Lambda()
        rho_psi = 0.5 * (psi_p * H0)**2 + V_full(psi, m_psi)

        # Friedmann: H^2 (1 + K/(a^2 H^2)) = 8piG/3 * rho_tot
        # H^2 = [8piG/3 * (rm + rr + rho_psi + rL)] - K/a^2
        rho_tot = rm + rr + rho_psi + rL
        H2 = 8*np.pi/3 * rho_tot - K_phys/a**2
        if H2 <= 0:
            return [0, 0]
        H = np.sqrt(H2)

        # Scalar field: psi'' + (3 + H'/H) psi' + V'/H^2 = xi*rm/H^2
        # where ' = d/dlna
        # H'/H = -(1/2)(3*rm + 4*rr + 3*psi_p^2*H^2) * 8piG/(3H^2)
        #          + K/(a^2*H^2)
        Hp_over_H = (-4*np.pi*(rm + 4*rr/3 + (psi_p*H)**2)
                     + K_phys/a**2) / H2

```

```

    dpsi = psi_p
    ddpsi = (-(3 + Hp_over_H) * psi_p
             - dV_full(psi, m_psi)/H2
             + xi * rm / H2)

    return [dpsi, ddpsi]

lna_span = [np.log(a_i), np.log(a_f)]
lna_eval = np.linspace(*lna_span, 5000)
y0 = [psi_i, 0.0]

sol = solve_ivp(system, lna_span, y0,
                 t_eval=lna_eval, rtol=1e-10, atol=1e-12,
                 method='RK45')

# Compute Omega_K at each point
a_arr = np.exp(sol.t)
psi_arr = sol.y[0]

Omega_K_arr = np.zeros_like(a_arr)
for i, a in enumerate(a_arr):
    rm = rho_m(a)
    rr = rho_r(a)
    rL = rho_Lambda()
    rho_psi = V_full(psi_arr[i], m_psi)
    H2 = 8*np.pi/3*(rm+rr+rho_psi+rL) - K_phys/a**2
    Omega_K_arr[i] = -K_phys / (a**2 * H2) if H2 > 0 else 0

return a_arr, Omega_K_arr, psi_arr

# === LCDM comparison (xi = 0) ===
def Omega_K_LCDM(a, K_phys):
    rm = rho_m(a)
    rr = rho_r(a)
    rL = rho_Lambda()
    H2 = 8*np.pi/3*(rm+rr+rL) - K_phys/a**2
    return -K_phys / (a**2 * H2) if H2 > 0 else 0

# === Run for three xi values ===
K_init = -0.01 * (1e-4)**2 * H0**2 # Omega_K,i = 0.01 at a=1e-4
for xi_val in [0.01, 0.05, 0.1]:
    a_arr, OK_arr, _ = evolve_system(xi_val, H0, K_init)
    OK_lcdm = np.array([Omega_K_LCDM(a, K_init) for a in a_arr])
    beta_eff = 1 - np.abs(OK_arr) / np.abs(OK_lcdm + 1e-30)

```



```
print(f"xi={xi_val}: |Omega_K(today)|={abs(OK_arr[-1]):.6f}")

# === Perturbation suppression at recombination ===
a_rec = 9.2e-4
k_star = 0.05 # Mpc^{-1}
m_psi_mpc = 2.2e-4 # H_0 in Mpc^{-1}
for xi_val in [1e-3, 0.01, 0.1, 1.0]:
    ratio = xi_val**2 * (m_psi_mpc * a_rec / k_star)**2
    print(f"xi={xi_val}: drho_psi/drho_m = {ratio:.2e}")
```

Peer Reviews and Revision History

First Round

Reviewer assessment. The reviewer found the paper to deliver on all three promissory notes carried from Paper 6: the mass hierarchy, CMB compatibility, and the full dynamical β . The mass hierarchy dissolution was judged the paper’s strongest conceptual contribution, with the condensed-matter parallel described as effective. The CMB compatibility analysis was found to be solid if semi-analytical. The dynamical β confirmation was praised as completing a proof chain building across three papers.

Five major concerns were raised: (1) the dissolution argument is philosophically effective but formally incomplete—Proposition 3.1 is asymmetric, with part (a) derived and part (b) assumed; (2) the ISW analysis is heuristic, not derived from the perturbation equations, yet is described as the “principal near-term observational signature”; (3) the numerical β solution lacks reproducibility details, uses the quadratic potential approximation at early times where it may fail, and risks circularity by imposing a β -dependent Ω_K evolution; (4) the temporal continuity of the screening field between inflation and late-time dynamics is not addressed; (5) the null predictions ($\mu = \Sigma = 1$, $c_T^2 = 1$) shared with Λ CDM create a falsifiability concern, and no lower bound on ξ is identified.

Six minor concerns were raised: (A) the cosmological coincidence problem ($m_\psi \sim H_0$) deserves more attention; (B) Table 1 conflates cosmic variance and instrumental noise; (C) the condensed-matter analogy should note the disanalogy in coupling-constant structure; (D) the QKDE citation [21] serves a programmatic rather than technical function; (E) unused reference to Kriger (2026) “Structural Genesis” should be removed; (F) the “Recommended Journal” section is unusual.

Verdict: Major revision required.

Revisions Made

1. Proposition 3.1 restated with explicit asymmetry (Concern 1). Part (b) now reads “*accommodates* a scalar excitation...which is *consistent with* the EFT framework, but is not yet derived from the effective action.” The proposition itself states that a first-principles derivation of m_ψ remains open. A new paragraph in §3.2 distinguishes the epistemic status of the two modes. The counterarguments include a direct response to the asymmetry objection.

2. ISW enhancement downgraded to heuristic estimate (Concern 2). Equation (22) is now explicitly labeled as a “heuristic scaling estimate, not a result derived from the linearized perturbation equations.” The subsection is retitled “The Integrated Sachs–Wolfe Effect: A Heuristic Estimate.” All language describing the ISW as the “principal near-term observational signature” or “prediction” has been replaced with “qualitative indication requiring numerical confirmation.” The Conclusions and abstract have been updated accordingly.

3. Numerical β analysis strengthened (Concern 3). Four changes: (a) The full Starobinsky potential replaces the quadratic approximation, with explicit discussion of why the quadratic fails at early times. (b) Ω_K is now *computed* from the Friedmann equation at each timestep, not evolved via a β -dependent equation; β_{eff} is *extracted* by comparison

with Λ CDM, eliminating circularity. (c) Results are shown for three parameter combinations ($\xi = 0.01, 0.05, 0.1$) in an expanded Table 2. (d) Proposition 7.1 is restated as an empirical result (“at all tested epochs and for all tested parameter values”) rather than a general monotonicity claim. The appendix code reflects the corrected numerical procedure.

4. Temporal continuity addressed (Concern 4). A new subsection §3.4 (“Temporal Continuity: The Screening Field Between Epochs”) explains that ψ is present in the action at all times: it is in its ground state during radiation domination (no source), responds quasi-statically during matter domination (saturated), and becomes dynamically active at late times. The transition is continuous and automatic, analogous to different excitation branches having different amplitudes at different epochs.

5. Lower bound on ξ and falsifiability (Concern 5). A new subsection §8.1 identifies a lower bound on ξ from the requirement that the screening mechanism function. The null predictions (34) are now displayed prominently with an explicit falsifiability statement. A “conjunction test” is described: the combination of $w \neq -1$, scale-dependent growth at $k \sim H_0$, and $\mu = \Sigma = 1$ is distinctive to the elastic vacuum.

6. Cosmological coincidence discussion expanded (Minor A). A new subsection in the Discussion (§9.1) compares the $m_\psi \sim H_0$ tuning with the Λ CDM tuning of Λ , arguing that the elastic vacuum tuning is milder ($O(1)$ coincidence vs. $O(10^{122})$ fine-tuning) but not absent.

7. Table 1 clarified (Minor B). The column headers now distinguish “Instrument noise” from “Cosmic variance,” with a caption explaining the distinction and noting that ISW-galaxy cross-correlations are not limited by cosmic variance.

8. Condensed-matter disanalogy noted (Minor C). §3.3 now states explicitly that in condensed matter, both modes arise from the same Hamiltonian with the same coupling constants, whereas in the gravitational case, the two modes are controlled by different parameters (M vs. ξ, m_ψ). The paragraph acknowledges that the analogy demonstrates genericity of large scale ratios but does not establish that the gravitational case is as benign as the condensed-matter case.

9. QKDE citation repositioned (Minor D). The Brown (2026) QKDE reference [21] is retained in the reference list as part of the programme’s intellectual context but is no longer cited alongside foundational references in the main text. It appears only in the bibliography.

10. Unused reference removed (Minor E). The reference to Kriger (2026), “Structural Genesis of Dynamical Architecture,” has been removed from the reference list, as it is not cited in the body of this paper. It will be cited in papers where it is used.

11. Recommended Journal section retained (Minor F). This section is part of the author’s standard manuscript template and will be removed upon journal submission if required by editorial policy.